

Article

Application of Overground Rock Film Mulching (ORFM) Technology in Karst Rocky Desertification Farmland: Improving Soil Moisture Environment and Crop Root Growth

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Abstract: Overground rock is a prominent feature of rocky desertification landscape in karst farmland; however, people often pay attention to their adverse effects, leaving their positive effects on ecohydrological processes and plant growth as rarely studied and utilized. In this study, the effects of overground rock film mulching (ORFM) on soil water flow behavior, soil water content and temporal and spatial heterogeneity were investigated through a dye tracer test and soil moisture measurement. Moreover, the effects of this technology on the root characteristics of crops (maize and broad bean) were analyzed. The results showed that ORFM treatment significantly increased soil water content and its spatio-temporal heterogeneity by preventing preferential flow at the rock–soil interface. It suggested that this practice can provide a more favorable soil moisture environment for crop growth, which was confirmed by the differences in root characteristics of crops (maize and broad bean) under different treatments in this study. It was found that ORFM treatment reduced the root radial extent of crops but increased the root biomass and root bifurcation rate, which are widely considered to be key factors in improving the efficiency of fine root absorption. Therefore, we believe that ORFM has great potential to improve the effective use of soil water and agricultural water management in karst areas, which is essential for sustainable agricultural development in the region.

Keywords: agricultural water conservation; karst farmland rocky desertification; overground rock film mulching; soil water flow characteristics; roots characteristics



Citation: Zhao, Z.; Zhang, J.; Liu, R. Application of Overground Rock Film Mulching (ORFM) Technology in Karst Rocky Desertification Farmland: Improving Soil Moisture Environment and Crop Root Growth. *Agronomy* **2024**, *14*, 1265. <https://doi.org/10.3390/agronomy14061265>

Academic Editor: Koffi Djaman

Received: 7 May 2024

Revised: 4 June 2024

Accepted: 11 June 2024

Published: 12 June 2024



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1. Introduction

In the mountainous and hilly areas of karst areas, due to unreasonable land use (such as over-cultivation) by human beings, the soil and vegetation have been destroyed, resulting in extensive soil erosion and rocky desertification in sloping farmland [1–3]. The typical characteristics of this area are scattered and discontinuous cultivated land, poor and shallow soil, and low land productivity [4–6]. Considering the compound effects of soil erosion and rocky desertification drought in this region, it is of great importance to analyze the surface soil hydrological process under this unique geological background and propose effective soil and water conservation measures for the reduction in soil obstacle factors, the improvement of basic land capacity, and the efficient utilization of water and fertilizer resources [7,8].

The hydrogeological structure of karst areas directly affects the availability and sustainability of water resources [9,10]. In southwest China, rainfall often occurs at the same time as high temperature, but the distribution of water resources is often very uneven [11,12]. The unique above-ground dual structure and porous media of bedrocks in karst areas lead to a rapid loss of soil water into shallow cracks, pipes and funnels [13,14]. Compared with

non-karst areas, this unique hydrological transport system leads to significantly different soil erosion processes and mechanisms [15–18], mainly due to shallow, discontinuous and limited soil layers and low water storage capacity [19–21]. Studies have shown that the soft and hard interface between rock and soil may become one of the key factors leading to soil and water loss [22–25], but its characteristics and its impact on soil water loss in sloping farmland are not fully understood [26,27]. At present, the research on soil erosion in karst areas involves many aspects, but due to the challenge of direct observation and the lack of practical research methods, the report on new measures of the prevention and control of soil erosion is limited [28–30].

Karst landforms, with their unique environmental characteristics, profoundly affect the water use and adaptation strategies of plants. Root characteristics can be taken as direct indicators of plant survival strategies [31–33]. The shallow soil layer in this area greatly restricts the vertical growth of roots, and plants adaptively spread roots horizontally and form simple fishtail branch structures to expand their root radial extent [34–37]. In general, this branching pattern could increase the length of the connection, so that it can access nutrients from a larger area [38–40]. With increasing drought severity, the average connection length of roots will increase, while the branching capacity decreases [41,42]. Although much progress has been made in the research of plant roots, the research of plant roots in karst areas is still insufficient. At present, most studies focus on non-karst areas, and the understanding of karst plant roots is still very shallow. In addition, reports on how soil and water conservation measures affect plant root characteristics in karst areas are extremely rare. The gaps in these research fields undoubtedly limit our comprehensive understanding of plant survival strategies in karst areas.

Overground rock in rocky desertification areas is a significant external factor of karst topography and is often regarded as an obstacle to improving land productivity. Despite widespread concern about their possible adverse effects on ecosystems, the fact remains that these rocks make a constructive contribution to the health and development of ecosystems. In proportion, rocks may play an important role in influencing water permeability and regulating soil water recharge [25,27]. Therefore, the in-depth study of the positive role of these rocks in the ecosystem will help us better understand the ecohydrological process of karst rocky desertification areas and provide a new strategy for ecological restoration and water resources management in this area. Recent studies have shown that overground rock film mulching (ORFM) can effectively prevent preferential flow at the rock–soil interface, prevent water loss and promote crop growth [43,44]. However, there is still a lack of relevant research reports on how ORFM specifically affects surface soil water and crop root characteristics in karst areas. In order to solve this problem, the flow behavior, water content and spatiotemporal heterogeneity of surface soil water in different treatment plots (i.e., film-mulched and non-mulched) were investigated. In addition, by combining field excavation measurements of plant root characteristics, we sought to gain a deeper understanding of the effects of ORFM on plant water use and adaptation mechanisms.

2. Materials and Methods

2.1. Study Area

Located in the hinterland of southwest China, Guizhou has a subtropical monsoon climate with an average annual rainfall of about 1300 mm and an average annual temperature of about 15 °C. Carbonate rocks are widely distributed, and the rocks are mainly composed of limestone and dolomite. Under the erosion of weathering and water flow, they form unique surface karst landforms, including karst funnels, karst depressions, karst basins, karst plains, peak clusters, and peak forests, etc. Guizhou has become the province with the most serious land rocky desertification problem in China, as the soil layer is shallow and discontinuous, and part of the bedrock is exposed to the surface. Rocky desertification causes poor soil quality and difficult-to-maintain soil moisture, which brings great troubles to agricultural production. Moreover, with the growth of the population and economic

development, the increasing demand for land resources also makes the already fragile ecological environment bear greater pressure.

The experimental field of this study is located in the Makan Village, Lvhu County, Qianxi City, Guizhou Province, which is located in a severe rocky desertification farmland area ($26^{\circ}57'23''$ N, $106^{\circ}6'21''$ E). Lvhu County is mainly mountainous, with an altitude between 1200 and 1510 m. The mean annual temperature is 13.8°C ; the mean annual rainfall is 1058 mm, most of which is concentrated in May to October. This is an area where the rocky desertification of sloping land is particularly serious. The area is characterized by its complex and diverse microgeomorphic landscape, displaying typical features of karst landforms. In this piece of farmland, rocks of different shapes are scattered among them (Figure 1). The soil type of the area is mainly brown and yellow limestone soil, and the specific soil properties are listed in Table 1.

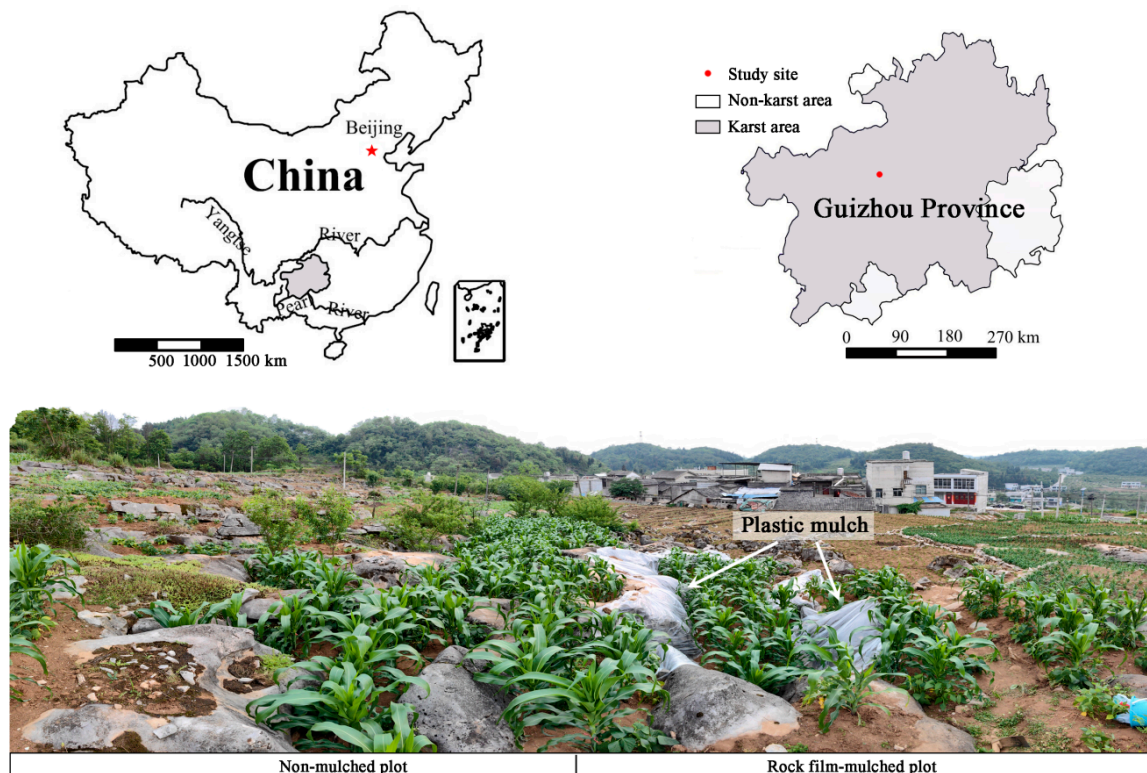


Figure 1. Location and environmental profile of the study area.

Table 1. Soil physical properties of the study site.

Soil Horizon	Depth (cm)	Soil Particle Composition (%)			Soil Texture (USDA System)	Bulk Density ($\text{g}\cdot\text{cm}^{-3}$)	Initial Water Content (% vol.)
		0.05~2 mm	0.002~0.05 mm	<0.002 mm			
A	0~15	17.33 ± 1.75	49.49 ± 1.08	33.17 ± 1.06	Silty clay loam	1.13 ± 0.02 b	11.07 ± 0.23 b
B	>15	7.39 ± 0.93	45.27 ± 0.60	47.34 ± 0.58	Silty clay	1.27 ± 0.02 a	14.18 ± 0.49 a

Notes: Data are expressed as the mean $\pm$ SE. $n = 9$. Different lowercase letters within rows indicate significant differences among soil horizons ($p < 0.05$). A: eluvial horizon; B: illuvial horizon.

2.2. Experimental Design

2.2.1. Application of ORFM

Before the mulching, brush, weeds and rock debris around the rocks were removed. Plastic film materials should have sufficient toughness and weathering resistance. Enough soil was prepared to hold the film in place and prevent it from being blown away or displaced by wind. Plastic film was placed over the rock surface to ensure close contact with the rock surface. For larger rocks, multiple sections of cover were required to ensure

the complete coverage of the entire rock surface. Regular inspections should be carried out to repair damaged or aged parts in a timely manner. The integrity of the film should be protected during maintenance to avoid human damage.

2.2.2. Dye Tracer Test

In order to study the effect of ORFM on soil water infiltration characteristics, six dye tracer tests were conducted in each sample plot, and bright blue FCF (4 g/L) was used as a dye tracer to simulate the flow and infiltration process of soil water. During each test, we performed the following steps: Before applying the dye, we removed weeds and debris from the floor to avoid interference with dyeing. A quantity of 6 L of the tracer solution was then sprayed onto the rock surface. This process lasted about 10 min to ensure the adequate penetration of the dye into the soil. After 24 h of the experiment, the vertical soil profile was excavated along the dyeing zone to observe the distribution of dye in the soil. Finally, dyeing depth, dyeing width, and dyeing coverage (in the profile range of 20 cm wide and 30 cm deep) were measured.

2.2.3. Soil Moisture Measurement

In the study area, two relatively uniform test plots were selected, each with a scale of 10 m × 10 m. One plot was treated with ORFM, and the other was treated without ORFM as the control group. The selected plots belong to moderate rocky desertification and are in a no-tillage state. The two plots were as similar as possible in terms of geology, slope orientation, vegetation cover, etc., in order to minimize the interference of non-experimental factors on the results. Before the rainy season (i.e., in February), we conducted a soil moisture measurement to assess the initial situation of soil moisture in the previous period. In the five months from May to September during the rainy season, one rainfall event was randomly selected every month, and the soil water content was measured at noon (11:00~14:00) on the first day and the first week after each rainfall. In each plot, a soil moisture meter (ML2-UM-3.0, Delta-T Devices Ltd., Cambridge, UK) was used to measure the surface soil volumetric moisture content, and measuring points were set every 1 m to form a grid distribution to cover the plots. For the points occupied by surface rocks and unable to measure soil water content, the mean imputation method was used to fill in the missing values.

2.2.4. Root Characteristic Measurement

The root characteristics of three crops (maize and broad bean) were studied by random complete block design. A representative rocky desertification field was selected for each crop as a test field. Each field was divided into three relatively uniform blocks, each of which was subdivided into two test plots and formed a control group of mulched and non-mulched plots (Figure 2). We ensured that the environmental conditions (terrain, soil, light conditions, etc.) in the region were relatively uniform, so as to reduce the interference of non-experimental factors on the results. We planted according to the crop planting code, ensuring that each crop in each field had the same planting density and growing conditions. During crop growth, necessary irrigation, fertilization and weed control measures were implemented to ensure normal growth. At the crop maturity stage, three crops with similar growth status were selected as replicate samples from each plot to measure root biomass, root distribution range and root bifurcation rate.

- Root Biomass

The above-ground part of the plant was removed in order to facilitate the collection of roots. With the crop as the center, we dug from the soil surface in the horizontal range of 30 cm × 30 cm and dug down layer by layer every 5 cm until the roots were completely collected. The collected plant root samples were then taken back to the lab, where they were cleaned to remove soil particles attached to the roots. Then, the cleaned root sample was placed in a drying oven and dried at a constant temperature of 80 °C to a fixed weight.

After drying, root samples were weighed to record the weight of each crop in different soil layers.

- Root Radial Extent

With the plant stem as the center, the excavation area was gradually expanded from top to bottom and from inside to outside. The horizontal distribution position of the roots in all directions was measured, and the root radial extent was mapped with a radar map on the coordinate paper. For root parts that penetrate rock cracks, grooves or other bedrock, complete root information should be obtained as far as possible using manual methods such as pry digging without damaging the root. We avoided damaging tree roots when digging. Measurements should be as accurate as possible to ensure the reliability of the results.

- Root Bifurcation Ratio

The outermost root was designated as a first-order root, and two roots of the same order formed a root of higher order. If the roots of different orders converged, the root of the higher order was taken as the order of the convergent root. The number of roots (N_i) of each order (i) was recorded and plotted with order (i) as the X-axis and $\lg N_i$ as the Y-axis. A straight line was obtained by linear regression, and the slope of this line was calculated. The anti-logarithm of the absolute value of slope of the regression line was taken as the total bifurcation ratio of the roots. The stepwise root bifurcation ratio ($R_i:R_{i+1}$) indicates the ratio of roots between adjacent steps, that is, $R_i:R_{i+1} = N_i:N_{i+1}$. Accuracy was crucial during the root classification and counting process to avoid the misclassification of the root order.

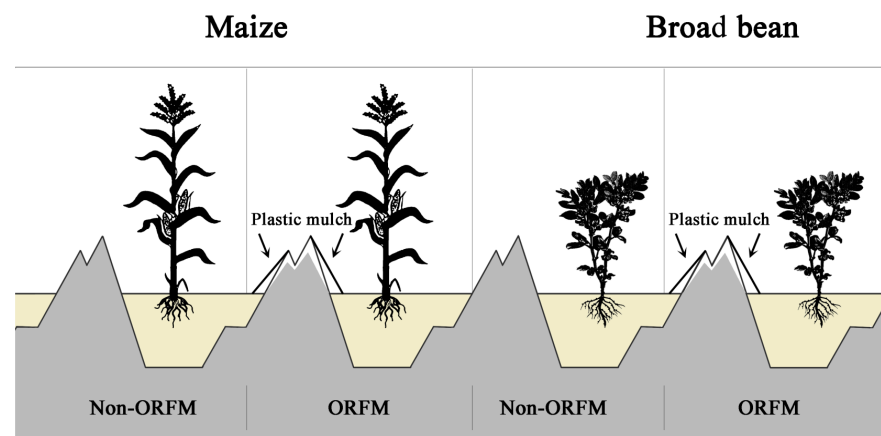


Figure 2. Schematic diagram of experimental design of the ORFM treatment. ORFM, overground rock film mulching.

3. Results

3.1. Soil Water Flow Behavior

There were significant differences in the infiltration behavior of dye solution under the two different treatment conditions (Figure 3 and Table 2). For instance, preferential flow dominated in the non-mulched plot, and the dye solution mainly penetrated vertically along the rock–soil interface. The maximum dyeing depth reached 30 cm, and the maximum dyeing width was about 10 cm, gradually decreasing with the increase in soil depth. By contrast, in the film-mulched plot, the preferential flow in the soil was not prominent. The dye solution did not penetrate directly along the rock–soil interface but was transferred to the surrounding soil and permeated downward in the form of matrix flow. The dyeing depth was less than 15 cm, and the dyeing width exceeded the measured width of the soil profile and decreased rapidly with soil depth.

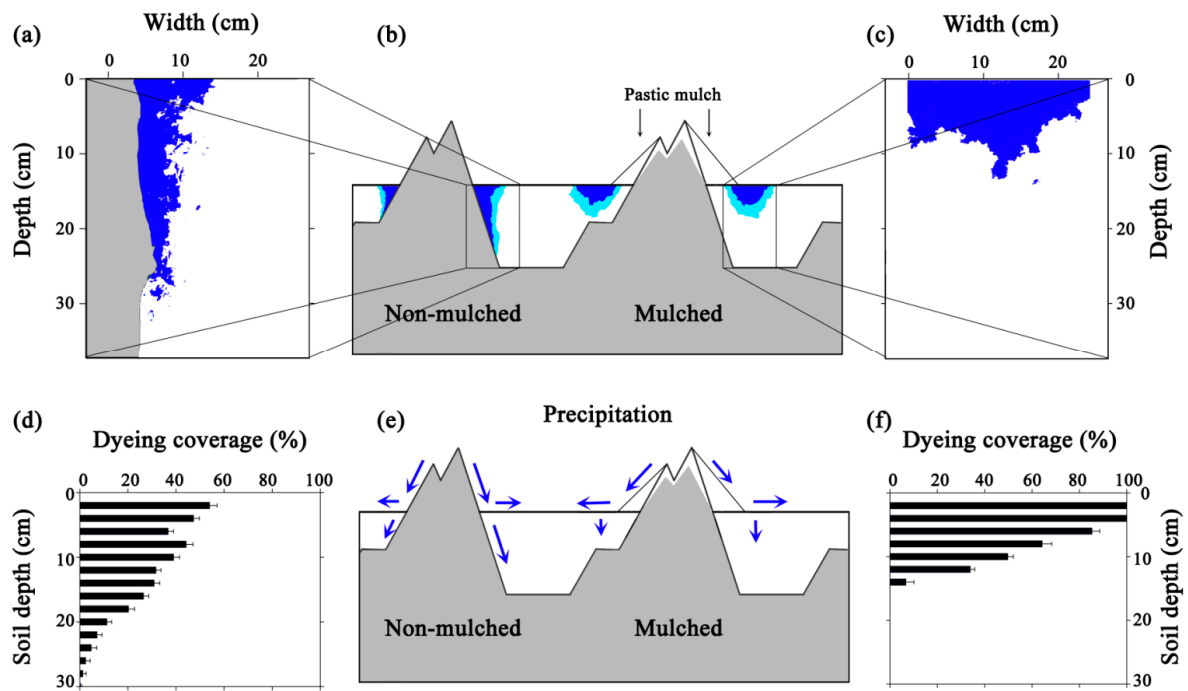


Figure 3. Infiltration patterns (b) in plots of the non-mulched (a) and the mulched (c). Dyeing coverage of the soil profiles in plots of the non-mulched (d) and the mulched (f). Flow behaviors of soil water in plots of the non-mulched and the mulched during rainfall (e).

Table 2. Characteristics of dyed soil profiles.

Treatments	Dyeing Width	Dyeing Depth	Dyeing Coverage
Mulched	24.97 ± 0.53 a	12.82 ± 0.57 b	29.33 ± 0.68 a
Non-mulched	10.80 ± 0.55 b	24.05 ± 1.30 a	23.86 ± 0.61 b

Notes: Data are expressed as the mean ± SE. $n = 6$. Different lowercase letters within rows indicate significant differences among treatments ($p < 0.05$).

3.2. Soil Water Content and Spatiotemporal Heterogeneity

Regardless of the ORFM treatment, the soil water content of the two plots was lower and similar at the beginning of the experiment. The CV_S of soil water content in both plots remained at a low level, indicating that the spatial distribution of soil water content is relatively uniform. From Figure 4, we can see that there was no significant local difference in soil water content in the two plots, which helped to evaluate the effect of ORFM treatment on soil water content. After the rains, however, the situation changed dramatically. On the first day and the first week after rain, the average soil water content of film-mulched plots was significantly higher than that of non-mulched plots (Figure 5). The CV_S of soil water content in mulched plots also increased significantly, showing more local areas of high water content (Figure 4). In addition, from the perspective of five rainfall events, the CV_T of soil water content after rain in the mulched plot was higher than that in the non-mulched plot (Table 3). Pearson correlation analysis showed that on the first day and the first week after rain, there was a significant correlation between the spatial distribution of soil water between different rainfall events, especially in the mulched plot (Figure 6).

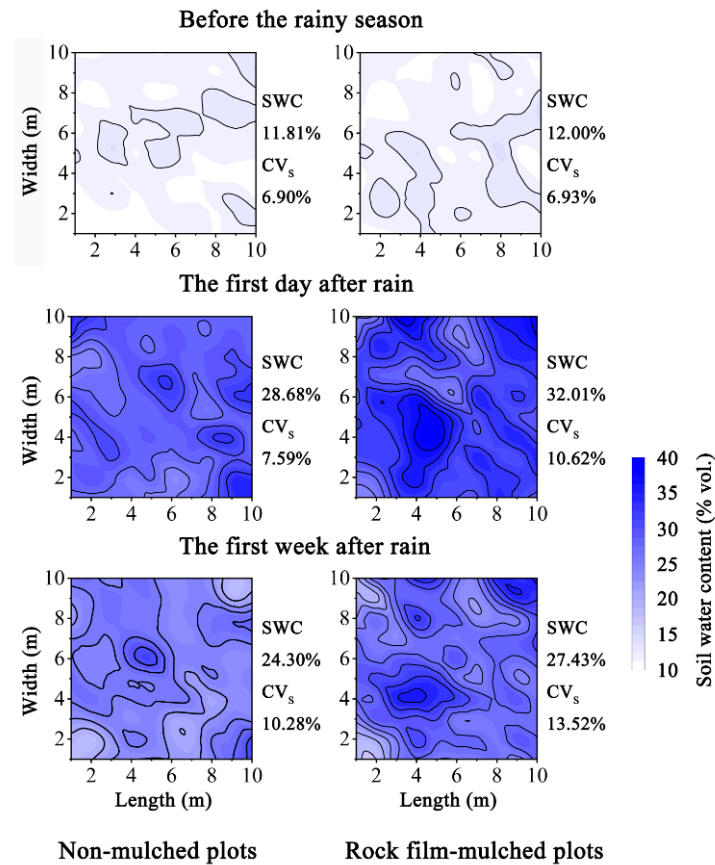


Figure 4. Spatial distribution of soil water content. SWC, Soil volumetric water content; CV_s, soil water content's coefficient of variation in space.

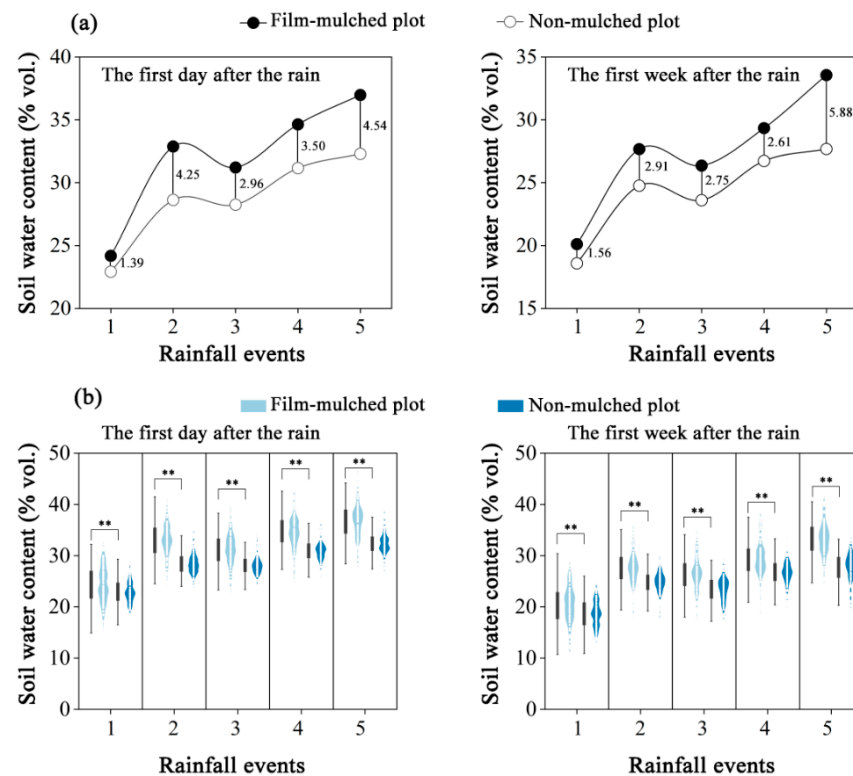


Figure 5. Dynamic change in soil water content (a) and its difference (b) between the rock film-mulched plot and the non-mulched plot. ** $p < 0.01$.

Table 3. Temporal heterogeneity of soil water content of five rainfall events.

Rainfall Events	Soil Water Content (% vol.)			
	The First Day after Rain		The First Week after Rain	
	Non-Mulched	Mulched	Non-Mulched	Mulched
1	22.92 ± 0.27 b	24.31 ± 0.38 a	18.69 ± 0.28 b	20.32 ± 0.42 a
2	28.63 ± 0.24 b	32.88 ± 0.36 a	24.76 ± 0.27 b	27.65 ± 0.39 a
3	28.25 ± 0.23 b	31.21 ± 0.35 a	23.61 ± 0.29 b	26.34 ± 0.38 a
4	31.16 ± 0.24 b	34.66 ± 0.37 a	26.73 ± 0.30 b	29.33 ± 0.41 a
5	32.44 ± 0.26 b	36.98 ± 0.38 a	27.68 ± 0.32 b	33.54 ± 0.40 a
CV _T (%)	12.77	15.02	14.48	17.55

Notes: Data are expressed as the mean ± SE. *n* = 100. Different lowercase letters within rows indicate significant differences among plot types (*p* < 0.01). CV_T, coefficient of variation in time.

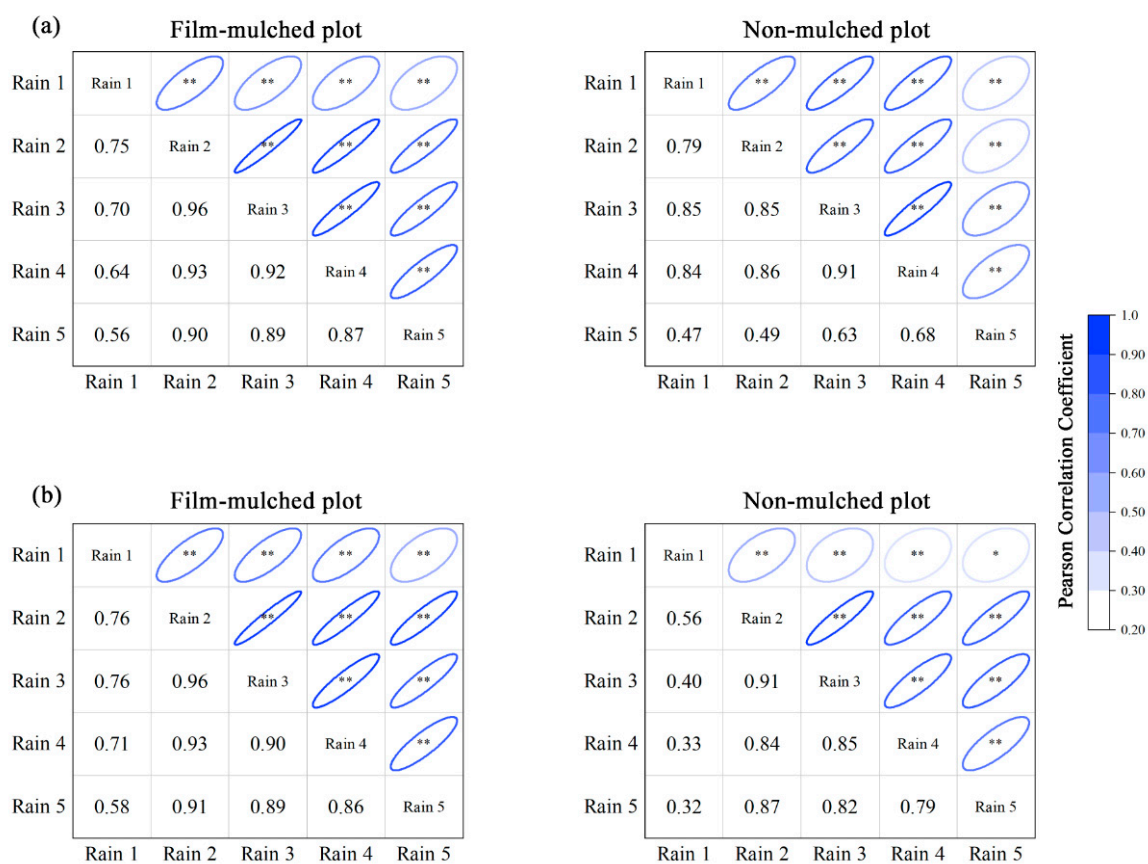


Figure 6. Pearson correlation coefficient of soil water spatial distribution between different rainfall events after the first day (a) and the first week (b). * *p* < 0.01; ** *p* < 0.001.

3.3. Root Characteristics

3.3.1. Root Biomass

Regardless of crop type, the total root biomass and root biomass in the shallow soil layer were generally greater in the rock film-mulched plots than in the non-mulched plots (Figure 7). The root biomass of maize in the deep soil of 15 cm depth showed a different trend: the root biomass of this soil layer in the non-mulched plots was greater than that in the mulched plots. Maize in >15 cm soil layers and broad bean in >20 cm were rarely found to have roots growing, so we did not collect roots here. For the vertical distribution of roots in soil, the root biomass of the three crops was mainly concentrated in the >15 cm soil layer and accounted for more than 90% of the total root biomass, which revealed the characteristic of a shallow distribution of crop roots in this area.

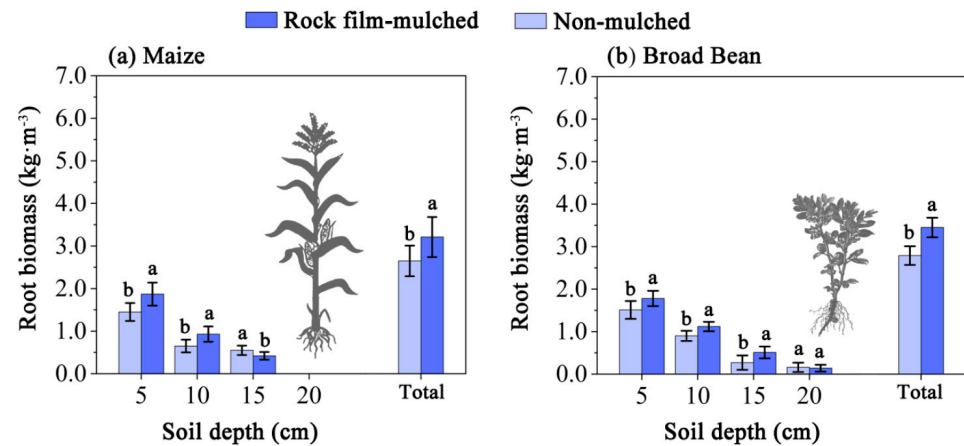


Figure 7. Comparison of crops root biomass under different treatments. Data are expressed as the mean $\pm$ SD. Different lowercase letters indicate significant differences between plot types ($p < 0.05$).

3.3.2. Root Radial Extent

We can observe from Figure 8 that the radial root extent of three crops differed between the two plots. Specifically, the average maximum root radial extent of maize was 13.87 cm and that of broad bean was 16.12 cm in the non-mulched plots, and the extension directions of roots also appeared quite uncertain with a diverse distribution pattern. Maize roots primarily exhibited a circle-shaped distribution, with growth directions involving multiple dimensions. In contrast, the roots of broad bean tended towards a irregular-shaped distribution, with most of the roots concentrated in some specific directions, and even exhibited unique growth patterns, such as wrapping around rocks and bending to seek for growth. In the mulched plots, the root radial extent of the crops showed significant trends. The maximum root radial extent of maize decreased to 11.76 cm and that of broad bean decreased to 14.49 cm, smaller than that in the non-mulched plots. Further, the roots showed a circle-shaped distribution, a more uniform and concentrated extent in all horizontal directions.

3.3.3. Root Bifurcation Ratio

As can be seen from Table 4, the total root bifurcation ratio and stepwise root bifurcation ratio of the three crops have significant differences between film-mulched and non-mulched plots. For example, compared with non-mulched plots, the total root bifurcation ratio of crops in mulched plots was significantly increased, and those of maize and broad bean increased by 17.84% and 7.40%, respectively. Likewise, the stepwise root bifurcation ratio of crops in mulched plots was also significantly increased, the R_1/R_2 of maize and broad bean increased by 12.26% and 9.25%, and the R_2/R_3 increased by 27.21% and 6.10%, respectively.

Table 4. Root bifurcation ratios of crops under different treatments.

Crops	Plots Type	R_b	R_i/R_{i+1}	
			R_1/R_2	R_2/R_3
Maize	Mulched	2.84 ± 0.02 a	1.65 ± 0.04 a	4.81 ± 0.06 a
	Non-mulched	2.41 ± 0.01 b	1.47 ± 0.03 b	3.78 ± 0.04 b
Broad bean	Mulched	1.74 ± 0.05 a	1.56 ± 0.03 a	1.95 ± 0.06 a
	Non-mulched	1.62 ± 0.03 b	1.41 ± 0.02 b	1.84 ± 0.05 b

Notes: Data are expressed as the mean $\pm$ SE. Different lowercase letters within rows indicate significant differences among plot types ($p < 0.05$). R_b , total root bifurcation ratio; R_i/R_{i+1} , stepwise root bifurcation ratio.

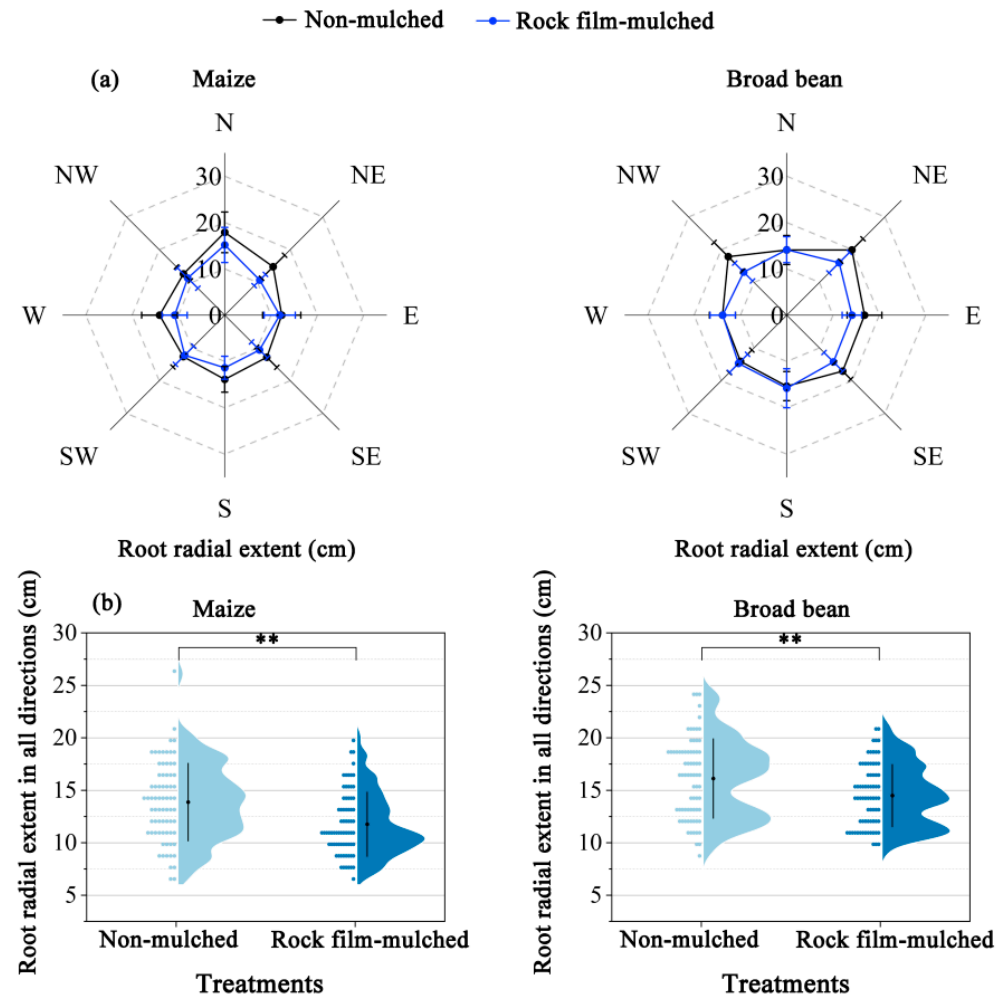


Figure 8. Root radial extent of crops under different treatments. (a) The spatial distribution of root in different directions. (b) The difference in root radial extent between different treatments. ** $p < 0.01$.

4. Discussion

4.1. Effect of ORFM on Soil Water

Compared with the plots without ORFM, the soil water content in the rock film-mulched plots increased from the first day to the first weeks after the rain (Figure 4 and Table 3), demonstrating that the ORFM has a significant effect on increasing soil water content. This improvement is undoubtedly a positive factor for the karst farmland ecosystem. In the area of serious rocky desertification, the funnel effect of aboveground rocks is particularly prominent [45], and the replenishment of surrounding soil by collecting rainwater has a significant effect on soil water content and plant growth [25]. The rocks not only receive atmospheric rainfall but also transfer nutrients to the surrounding soil and plants, thereby improving soil quality [46]. Although there are channels for vertical water leakage at the soil–rock interface, resulting in some water loss during the rainfall–runoff process [23,25,47], the application of ORFM effectively addresses this issue. Furthermore, a recent study has further shown that by blocking preferential flow at the soil–rock interface, ORFM significantly enhances the water-collecting capacity of rocks to enable more rainwater to be absorbed and utilized by plants [43,44], meaning this practice not only optimizes soil water management but also improves the efficiency of water resource utilization.

The unique geomorphological features of karst areas inherently confer strong variability to soil moisture for many reasons, and one of them was a differential water-collecting capacity of overground rocks [48]. In this study, ORFM not only increased soil water content quantitatively but also significantly enhanced the spatial and temporal variability of soil

water content in karst croplands (Figure 4 and Table 3). The increase in variability was also attributed to the enhanced water-collecting capacity of rocks by applying ORFM, thereby enlarging the influence of topography and rainfall on the soil water. ORFM mitigates soil water leakage and enables a wider range of soil patches to benefit from rocks, which further increases the heterogeneity of spatial patterns and temporal dynamics of soil water. The strong correlation of soil water spatial pattern between different rainfall events confirmed the importance of karst topography for rainfall redistribution (Figure 6). This finding proves that ORFM can help differentiate water conditions between soil patches, thereby providing more suitable habitats for maintaining biodiversity in karst ecosystems.

4.2. Effects of ORFM on Root Characteristics

This study showed that crop root biomass in the rock film-mulched plots was significantly higher than that in the non-mulched plots (Figure 7), indicating that ORFM, by increasing soil moisture, could promote root growth. Many studies have reported that, to some extent, root biomass is positively correlated with soil moisture [49–51]. Proper soil water content maintains the balance between soil aeration and water supply, which can create a very favorable environment for root growth. However, high soil water content can also adversely affect plant root growth. This is because high soil moisture reduces soil aeration to inhibit root respiration and also dilutes nutrients in the soil to reduce their availability. Therefore, in agricultural production, it is particularly important to promote crop growth by properly adjusting soil water content. Through this study, we believe that ORFM can optimize the soil water content in karst farmland and help the soil environment maintain a better moisture state.

ORFM concentrated the spatial distribution of crop roots (Figure 8). Plant roots seek water by reaching larger areas in arid environments [52]; however, in the karst soil-limited environment where the root's deep penetration is restricted, plants always adapt by forming large, radially extended, coarse root systems [35,53,54]. This not only helps the plant stabilize the soil better but also improves the plant's ability to obtain water and nutrients. It was found that in the non-mulched plots with low soil moisture, the roots of the three crops showed obvious multi-directivity and non-uniformity. In extreme cases, the root system may extend along the soil–rock interface or even through cracks in the rock to seek more space for expansion. In contrast, in the mulched plots with higher soil moisture, the root distribution was more concentrated and uniform. Some studies have confirmed that moist soil is more conducive to the growth of fine roots to improve absorption efficiency [52,53]. Therefore, it can be inferred that ORFM can concentrate plant roots and narrow the radial range of roots by improving soil water conditions, thus helping to improve the utilization efficiency of soil resources by plants.

ORFM increased the root bifurcation ratio of crops (Table 4). Root branching influences root's growth, distribution and absorption capacity [55]. In karst areas, plants generally adopt fishtail branching to cope with multiple environmental pressures. Specifically, by reducing the number of branches and internal competition within the root system while expanding the distribution range of roots over long distances, they maximize their access and utilization of soil resources. Moist soil prompts plants to increase their root branching [52,53], and ORFM, by increasing soil water content, was found to increase the total and stepwise root bifurcation ratio of crops in the study. Despite reducing root radial extent, it holds advantages in nutrient-rich habitats, since it increases the contact surface area between the roots and the soil and facilitates plants to absorb water and nutrients efficiently.

4.3. Application of ORFM

This study confirmed that, by preventing preferential flow at the rock–soil interface, ORFM increased soil water content and its spatial–temporal heterogeneity and influenced the root biomass, root radial extent and root branching of crops (Figure 9). ORFM was expected to increase the productivity of karst farmland ecosystems, considering its contribu-

tion to ensuring the water required for crop growth and enhancing root absorption capacity. In addition, the overuse of fertilizers and pesticides poses a threat to groundwater [56]. ORFM by blocking soil preferential flow is likely to reduce the risk of the downward infiltration of these pollutants, thereby protecting groundwater from contamination, which is beneficial to the agricultural environment health of karst farmland.

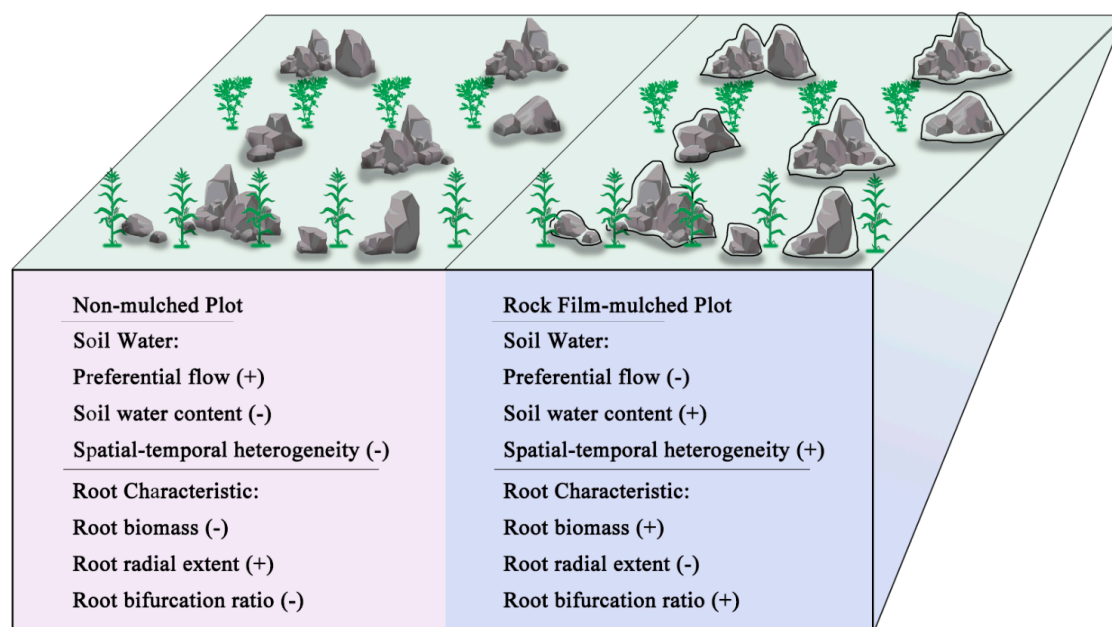


Figure 9. Effects of overground rock film mulching on the spatial-temporal heterogeneity of soil water and root characteristics of crops. “+” denotes positive effect; “−” denotes negative effect.

In practice, in order to achieve the best implementation effect, it is often necessary to choose suitable soil and water conservation technology according to the specific geological environment. Especially in karst areas, complex terrain and abundant surface rocks affect agricultural production activities. It is therefore essential to tailor soil and water conservation techniques to soil–rock microgeomorphic units. The essence of ORFM is to cover the overground rock surface to enhance the rock’s ability to redistribute rainfall to the surrounding soil. This practice as an assistive technology cannot completely replace other water conservation measures. Finally, ORFM is easy to operate and relatively low cost, which is conducive to its wide application and promotion, but the quality and environmental friendliness of the covering material must be considered to avoid secondary pollution.

5. Conclusions

Overground rock is a prominent feature of rocky desertification landscape in karst area. However, people often pay attention to its adverse effects on farmland ecosystems, while its positive effects on ecohydrological processes and plant growth are rarely studied and utilized. This study analyzed the impact of ORFM on soil water flow behavior, soil water content in croplands and its spatial-temporal heterogeneity, and the root characteristics of crops under different treatments were investigated. It was found that the application of ORFM significantly increased soil water content as well as its spatial-temporal heterogeneity by preventing preferential flow at the rock–soil interface, which suggested that ORFM can help improve the growing environment for crops by creating more suitable soil conditions. Additionally, it was found that ORFM decreased the root radial extent of crops but increased the root biomass and root bifurcation ratio, indicating that this practice favors fine root growth to improve absorption efficiency.

ORFM, as an efficient soil improvement technology, can show great application prospects in karst areas and provides useful guiding value for land use and soil water management. In the future, we need to accurately evaluate the actual benefits and potential

limitations of ORFM to soil water in rocky desertification areas and give a full picture of its advantages and long-term effects through scientific design and application, so as to contribute to improving the efficiency of land resource utilization and promoting the sustainable development of karst ecosystems.

Author Contributions: Conceptualization, Z.Z.; methodology, Z.Z. and J.Z.; software, Z.Z. and J.Z.; validation, Z.Z.; formal analysis, Z.Z.; investigation, Z.Z. and R.L.; resources, Z.Z.; data curation, Z.Z. and J.Z.; writing—original draft preparation, Z.Z.; writing—review and editing, Z.Z.; project administration, Z.Z.; funding acquisition, Z.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Guizhou Provincial Basic Research Program (Natural Science) (grant number QKHJC-ZK [2023]YB281) and the Guizhou Province Colleges and Universities Young Science and Technology Talents Growth Projects (grant number QJHKY [2022]296).

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Acknowledgments: All of the authors especially appreciate the experimental equipment provided by the Guizhou Provincial Key Laboratory of Geographic State Monitoring of Watershed, School of Geography and Resources, Guizhou Education University. We thank the anonymous reviewers for their valuable comments.

Conflicts of Interest: The authors declare no conflicts of interest.

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